

C Language Self Learning Program

REVISION HISTORY:

Revision	Date	Description	Author	Reviewed By	Approved By
1.0					

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1. Fundamental C Language Concepts

1.1 Chapter 1: Introduction to C Language

- Overview and Historical Background
- Structure of a C Program
- Compilers and Interpreters

1.2 Chapter 2: Basic Syntax and Data Types

- Variables and Constants
- Data Types: int, float, char
- Input and Output Functions

1.3 Chapter 3: Control Flow Statements

- Conditional Statements (if, else, switch)
- Looping Statements (for, while, do-while)
- Break and Continue Statements

1.4 Chapter 4: Functions and Scope

- Function Definition and Declaration
- Function Parameters and Return Values
- Scope of Variables (Local and Global)

1.5 Chapter 5: Arrays and Strings

- Array Declaration and Initialization
- Multi-dimensional Arrays
- String Handling Functions

1.6 Chapter 6: Pointers and Memory Management

- Introduction to Pointers
- Dynamic Memory Allocation (malloc, free)
- Pointer Arithmetic

1.7 Chapter 7: Structures and Unions

- Defining Structures and Unions
- Accessing Structure Members
- Nested Structures and Unions

1.8 Chapter 8: File Handling in C

- Opening, Reading, and Writing Files
- File Pointers and Modes
- Error Handling with Files

1.9 Chapter 9: Embedded C

- Bit manipulation/swapping
- Linked list
- volatile
- const volatile
- AND OR operations
- union vs struct

2. Coding Guidelines

2.1 Naming Conventions:

- Use Meaningful and Descriptive Names.
- Follow a Consistent Naming Convention.

2.2 Indentation and Formatting:

- Maintain Consistent Indentation.
- Keep Code Readable with Clear Formatting.

2.3 Comments:

- Add Comments for Complex Logic or Explanations.
- Update Comments During Code Modifications.

2.4 Modularization:

- Break Down Code into Modular Functions.
- Avoid Lengthy Functions for Readability.

2.5 Error Handling:

- Check for Errors and Provide Clear Error Messages.
- Gracefully Handle Exceptions.

2.6 Memory Management:

- Properly Allocate and Deallocate Memory.
- Avoid Memory Leaks.

2.7 Consistency:

- Adhere to Consistent Coding Styles and Conventions.
- Align with Project or Team Guidelines.

2.8 Testing:

- Conduct Thorough Testing Before Deployment.
- Consider Edge Cases in Testing.

3.Reference Links

- <https://www.geeksforgeeks.org/c-programming-for-beginners-a-20-day-curriculum/>
- <https://www.programiz.com/c-programming>
- <https://www.youtube.com/watch?v=0Sg6QHmIFJE&list=PL98qAXLA6aftD9ZInjpLhdQAOfI8xIB6e&index=31>
- <https://www.tutorialspoint.com/cprogramming/index.htm>
- <https://www.hackerrank.com/domains/c>
- <https://exercism.org/tracks/c>
- <https://www.codechef.com/learn/course/c>
- https://capgemini.sharepoint.com/:f:/r/sites/HRWorkspace/CGEngg_TTDL/Shared%20Documents/Nalanda/C?csf=1&web=1&e=hyl9hw
- [Coding Guidelines 1.3 \(2\).pdf](#)